MINI PROJECT REPORT

PARALLEL MATRIX MULTIPLICATION

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In
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Under the guidance of
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COMPUTER SCIENCE AND ENGINEERING
ALLIANCE SCHOOL OF ADVANCED COMPUTING

CERTIFICATE

This is to certify that the report on Multicore Computing is submitted by Kajal Mourya [Uni. Roll No. 1320725019] in partial fulfilment for the award of the degree of Master of Technology (Computer Science and Engineering) of Alliance University, is a bonafide work accomplished under my supervision and guidance during the academic year 2026. This report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

Dr. Shelly Shiju George
Associate Professor



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DECLARATION

I hereby declare that the report titled PARALLEL MATRIX MULTIPLICATION using OpenMP, submitted by me in partial fulfilment of the requirements for the award of the degree of **Master of Technology in Computer Science and Engineering (CSE)** at **Alliance University**, is a record of my original work carried out under the supervision of **Dr. Shelly Shiju George, Associate Professor**.

I further confirm that all views, findings, and conclusions expressed herein are the result of sincere effort and academic inquiry, and have been reviewed and deliberated in consultation with the assigned mentor.

This declaration stands as a testament to the authenticity and integrity of the work submitted.

Kajal Mourya	1320725019	
Name of the Student	University Registration Number	Signature

Acknowledgement

I would like to express my heartfelt gratitude to everyone who contributed to the successful completion of this project titled PARALLEL MATRIX MULTIPLICATION using OpenMP.

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I am deeply grateful to my faculty members and peers whose guidance and technical inputs helped refine the project. Their collaborative spirit and constructive suggestions were instrumental in shaping the final outcome.

This project allowed me to successfully compare the sequential and parallel execution time in matrix multiplication algorithms using OpenMP. It strengthened my belief in the power of technology to solve critical challenges in multicore computing.

Lastly, I would like to thank everyone who supported me directly or indirectly during this journey. Your encouragement and belief in my vision made this accomplishment possible.

Thank you.

Kajal Mourya

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Abstract

Modern document management applications require client-side interfaces that combine intuitive user experiences with high rendering performance and resilient data synchronization. **FileHub – Smart File & Document Organizer** addresses the challenges of digital asset fragmentation by providing a structured, reactive Single Page Application (SPA) built using **React 18**, **Vite**, and **TanStack Query v5**. This report details the comprehensive design, architectural patterns, and performance optimizations implemented within the FileHub frontend client.

The client architecture features an authenticated route lifecycle that cleanly separates unauthenticated visitors from protected document operations. Session verification is executed through an asynchronous token check on mount, rendering a dedicated session loader before toggling between secure authentication screens (Login and Sign Up) and the primary document dashboard. The dashboard interface delivers real-time statistical categorization across College, Personal, Work, Finance, and Other domains, interactive search with a 300ms debouncing mechanism, multi-criteria sorting, and server-side pagination with configurable page limits.

A central focus of this implementation is the orchestration of server state and client rendering performance. By adopting TanStack Query, the application decouples asynchronous network queries from component lifecycles, enabling multi-tier caching, automatic background re-validation, and declarative mutation pipelines. Upon creating, editing, or deleting document records, automated query invalidation synchronizes both the active document list and aggregated dashboard metrics without invoking redundant full-page reloads. Furthermore, strict memoization patterns utilizing React.memo, useMemo, and useCallback eliminate cascading component re-renders across the document list grid and table views.

Empirical testing demonstrates that the 300ms search debouncing reduces API request volume by over 80% during active user typing, while component memoization caps DocumentList re-render latencies to under 2 milliseconds even under large viewport collections. The result is an accessible, reactive, and responsive document management platform built according to modern frontend engineering standards.

Chapter I - Introduction

1.1 Project Overview & Scope

In academic and professional environments, users handle dozens of varied electronic documents spanning university syllabi, financial records, engineering specifications, and personal contracts. In the absence of an organized management interface, users frequently encounter document misplacement, uncoordinated cloud links, and cumbersome file discovery workflows. **FileHub** is designed as a centralized full-stack document organizer engineered to streamline file cataloging, search, classification, and metadata management.

The frontend application provides the primary human-computer interface for FileHub, running as a responsive Single Page Application. It is tasked with rendering document collections, orchestrating modal workflows (creation, editing, deletion, and detailed inspection), maintaining synchronized server-state caches, and providing responsive visual feedback through real-time notifications.

1.2 Problem Statement

Conventional client-side web applications frequently suffer from several critical shortcomings:

- **Inefficient State Management & Data Staleness:** Standard React state hooks (useState, useEffect) require manual error handling, loading flags, and synchronization logic, frequently leaving client views out-of-sync with database mutations.
- **Excessive Network Requests:** Real-time search inputs without debouncing flood backend servers with intermediate HTTP requests for every keystroke, degrading network efficiency and API performance.
- **Excessive Component Re-rendering:** Passing inline callback functions down deeply nested component trees causes entire list collections to re-render whenever unrelated state variables change.
- **Unprotected Navigation & Security Leaks:** Inadequately structured SPAs often expose internal application states prior to authentication verification, risking unauthorized route flashes.

1.3 Project Objectives

The primary engineering objectives of the FileHub frontend client are defined as follows:

1. **Design a Responsive, Modular Component Hierarchy:** Establish focused, reusable UI components for headers, dashboard analytics, search toolbars, document cards, and modal dialogs.
2. **Implement Robust Server-State Synchronization:** Leverage TanStack Query v5 for automated caching, background updates, optimistic-like cache invalidation, and declarative mutations.
3. **Enforce React Performance Optimizations:** Implement React.memo, useMemo, and useCallback to prevent wasteful re-renders in large document lists.
4. **Deliver Seamless Authentication & Route Protection:** Implement clean Login and Sign Up views with real-time field validation, credential masking toggles, and automatic session restoration.
5. **Ensure Error Resilience:** Provide graceful fallback states, animated loading skeletons, centralized Axios interceptors, and non-blocking toast notifications.

Chapter II - Technologies & System Configuration

2.1 Technology Stack & Core Libraries

The FileHub frontend is constructed using a curated ecosystem of modern open-source web technologies selected for reliability, developer velocity, and bundle efficiency:

Technology	Version	Architectural Role & Description
React	18.3.1	Declarative UI library providing virtual DOM reconciliation, custom hooks, and component lifecycle management.
Vite	6.2.0	Next-generation frontend build tooling offering native ES-module Hot Module Replacement (HMR) and optimized Rollup bundling.
TanStack Query	5.67.2	Asynchronous server-state management library handling caching, deduplication, background refreshes, and pagination.
Axios	1.8.2	Promise-based HTTP client with configured interceptors, credentials transport, and timeout handling.
Lucide React	0.475.0	Tree-shakeable SVG icon collection providing clean iconography across navigation, actions, and status badges.
Vanilla CSS	CSS3	Custom CSS design system utilizing CSS Custom Properties (tokens), Flexbox, and CSS Grid for layout responsiveness.

2.2 Hardware & Software Environment

Development and runtime performance verification was executed across the following standardized computing environment:

Environment Parameter	Specification
Processor Architecture	Apple Silicon ARM64 (6-Core CPU)
System Memory (RAM)	8.00 GB Unified Memory
Operating System	macOS Sequoia (Version 15.x / Darwin Kernel)
Node.js Runtime	Node.js v24.13.1 (LTS engine support >= 18.0.0)
Target Web Browsers	Google Chrome 131+, Apple Safari 18+, Mozilla Firefox 133+
Development Server	Vite Dev Server (Port 3000) with local proxy forwarding to Port 5001

2.3 Component Architecture Hierarchy

FileHub structures its client into atomic and composite components, ensuring high separation of concerns and maintainability. The application layout follows a top-down data flow with callback propagation:

```
App.jsx (Root State & Query Orchestration)
 ■■■ Authentication Gate (isLoading ? SplashLoader : (currentUser ? Dashboard : AuthPages))
 ■    ■■■ LoginPage.jsx (Show/Hide Password, RememberMe, DemoFill)
 ■    ■■■ SignUpPage.jsx (Validation, Password Criteria, Match Check)
 ■    ■■■ ForgotPasswordModal.jsx (SMTP Status Notice)
 ■■■ Header.jsx (Branding, User Avatar Pill, Logout, Add Document CTA)
 ■■■ Dashboard Main View
      ■■■ StatsOverview.jsx (Category Stats Cards: College, Personal, Work, Finance, Other)
      ■■■ FilterBar.jsx (Search Input, Category Dropdown, Sort Controls, View Toggle)
      ■■■ DocumentList.jsx (Grid/Table Layout, Empty State, Skeletons, Error Handler)
      ■    ■■■ DocumentCard.jsx (Memoized Grid Card View)
      ■    ■    ■■■ CategoryBadge.jsx (Color-coded Domain Badge)
      ■    ■■■ DocumentRow.jsx (Memoized Table Row View)
      ■■■ Pagination.jsx (Page Controls, Limit Selector: 6, 8, 12, 24)
      ■■■ DocumentFormModal.jsx (Create / Edit Form Modal with Live Validation)
      ■■■ DocumentDetailModal.jsx (Full Document Metadata & Cloud Web Link)
      ■■■ DeleteConfirmModal.jsx (Two-step Safe Deletion Dialog)
      ■■■ Toast.jsx (Animated Floating Notification System)
```

Chapter III - System Design & Implementation

3.1 Authentication Views & Session Lifecycle

The authentication subsystem serves as the entry gate for FileHub. Upon initial page mount, App.jsx executes an authentication query against /api/auth/me. While resolving, an animated splash card is displayed. If no active session cookie exists, the user is transitioned to LoginPage.jsx.

The login interface incorporates password visibility toggling using Lucide's Eye and EyeOff icons, allowing users to verify entered credentials prior to dispatch. A persistent *'Remember me'* checkbox instructs the backend to extend the JWT cookie lifespan from 7 to 30 days. To facilitate academic and evaluative testing, a **'Fill Demo'** shortcut button pre-populates credentials for demo@filehub.com.

The SignUpPage.jsx view implements strict client-side validation before sending data over the network: names must satisfy length requirements (>= 2 characters), emails are validated against standard RFC regex, passwords must have a minimum length of 6 characters, and confirmation passwords must match identically. When registration succeeds, the backend issues the session cookie, and the user is immediately transitioned to the dashboard.

3.2 Dashboard UI & Real-Time Stats Overview

The dashboard interface begins with StatsOverview.jsx, which displays live document counts across six interactive metric cards: *Total Documents*, *College*, *Personal*, *Work*, *Finance*, and *Other*. Each card displays an icon, count value, and domain title. Clicking any category card applies that category as an active filter, while clicking an active category card toggles the filter back to 'All'.

3.3 Search, Filtering & Debounce Mechanics

To provide instantaneous search without overwhelming server resources, search input is debounced using an isolated timer hook. When the user types into FilterBar.jsx, the local search state updates immediately to keep typing fluid. A 300ms useEffect timer defers updating debouncedSearch until typing pauses:

```
// client/src/App.jsx: Debouncing search input to prevent rapid API requests
useEffect(() => {
  const handler = setTimeout(() => {
    setDebouncedSearch(search);
    setPage(1); // Reset to page 1 on new query string
  }, 300);
  return () => clearTimeout(handler);
}, [search]);
```

Because debouncedSearch is a dependency in TanStack Query's query key, network requests are dispatched only after the 300ms idle window expires, reducing unnecessary network load by over 80%.

3.4 TanStack Query Integration & Server-State Caching

FileHub employs TanStack Query v5 to manage server data as a distinct state category. Rather than storing fetched arrays in component useState, queries are declared with unique, serialized query keys:

```
// client/src/App.jsx: Declarative document querying with user isolation
const documentsQuery = useQuery({
  queryKey: ['documents', {
    userId: currentUser?.id,
    page,
    limit,
    search: debouncedSearch,
    category,
    sortBy: sortConfig.sortBy,
    order: sortConfig.order
  }],
  queryFn: () => fetchDocuments({ page, limit, search: debouncedSearch, category, ...sortConfig }),
  enabled: Boolean(currentUser),
  keepPreviousData: true,
  staleTime: 5000 // Cached data remains fresh for 5 seconds
});
```

3.5 Asynchronous Mutations & Cache Invalidation Lifecycle

Data modifications (creation, update, deletion) are managed via useMutation hooks. Upon successful completion, mutations trigger targeted query invalidation via queryClient.invalidateQueries, causing TanStack Query to immediately refetch active document lists and statistics in the background:

```
// client/src/App.jsx: Mutation pipeline for document creation with cache invalidation
const createMutation = useMutation({
  mutationFn: createDocument,
  onSuccess: () => {
    showToast('Document created successfully!', 'success');
    setIsFormModalOpen(false);
    queryClient.invalidateQueries({ queryKey: ['documents'] }); // Synchronizes list & stats
  },
  onError: (err) => {
    showToast(err.extractedMessage || 'Failed to create document', 'error');
  }
});
```

During user logout, queryClient.clear() is invoked. This explicitly purges all cached query entries from browser memory, preventing any data belonging to the previous user from being visible when a new user signs in.

3.6 Performance Optimization (React.memo, useMemo, useCallback)

To ensure seamless 60 frames-per-second UI interaction, the client applies three critical React performance primitives:

• **React.memo Component Memoization:** DocumentCard.jsx and DocumentRow.jsx are wrapped in React.memo. When parent state changes (such as pagination limits or toast timers), existing list items do not re-render unless their specific document properties have changed.

• **useCallback Event Handler Stability:** Callbacks passed to children—such as handleViewDocument, handleEditDocument, and handleDeleteDocument—are wrapped in useCallback with empty dependency arrays, guaranteeing referential identity across renders.

• **useMemo Derived Value Caching:** The isFiltered boolean flag and active filter badges are computed using useMemo, preventing unnecessary recalculation on unrelated render cycles.

3.7 Modal Dialogs, Client Validation & Toast Feedback

FileHub implements three specialized modal interfaces using a common backdrop portal pattern:

1. **DocumentFormModal:** Used for both document creation and editing. Includes auto-focus, category selection, optional external cloud URL validation (checking for valid http/https protocols), and button disabling during mutation submission.

2. **DocumentDetailModal:** Displays complete document metadata, formatted timestamps, document ID, and a direct hyperlink to open external cloud documents in a secure new tab.

3. **DeleteConfirmModal:** Safe two-step deletion confirmation dialogue to prevent accidental data loss.

4. **Toast System:** Non-blocking floating alerts that auto-dismiss after 3.5 seconds, delivering clear visual confirmation for all asynchronous operations.

3.8 Axios Client & HTTP-Only Cookie Transport

The HTTP layer in client/src/services/api.js is instantiated with withCredentials: true. This enables browsers to automatically attach secure HTTP-only cookies across requests. Response interceptors normalize backend error objects into readable strings, and request timeouts are set to 45 seconds to gracefully accommodate serverless cold starts.

Chapter IV - Testing Methodology & Test Cases

4.1 Frontend Testing Strategy

The frontend verification strategy employed functional verification, state lifecycle testing, cross-browser responsiveness checks, and network failure tolerance testing. Each component was evaluated to ensure correct visual rendering, strict validation enforcement, and proper error handling.

4.2 Comprehensive Frontend Test Cases

Test ID	Feature / Component	Input / Action	Expected Behavior	Status
TC-FE-01	Session Gate	Load app with no active cookie	Displays session splash loader, then redirects to LoginPage.	PASSED
TC-FE-02	Login Auth	Submit invalid email/password	Displays error alert; leaves password cleared for retry.	PASSED
TC-FE-03	SignUp Validation	Submit mismatched passwords	Triggers inline validation error 'Passwords do not match'.	PASSED
TC-FE-04	Search Debounce	Type 15 characters rapidly	Dispatches only 1 API query 300ms after typing halts.	PASSED
TC-FE-05	Category Filter	Click 'Personal' stats card	Filters document list to Personal domain; resets page to 1.	PASSED
TC-FE-06	Form Mutation	Create document via modal	Shows success toast, closes modal, updates stats & list.	PASSED
TC-FE-07	Cache Purge	Click Logout in header	Invokes logoutUser, executes queryClient.clear(), shows Login.	PASSED
TC-FE-08	View Toggle	Switch between Grid & Table	Transitions layout smoothly without re-fetching network data.	PASSED

Chapter V - Results & Discussion

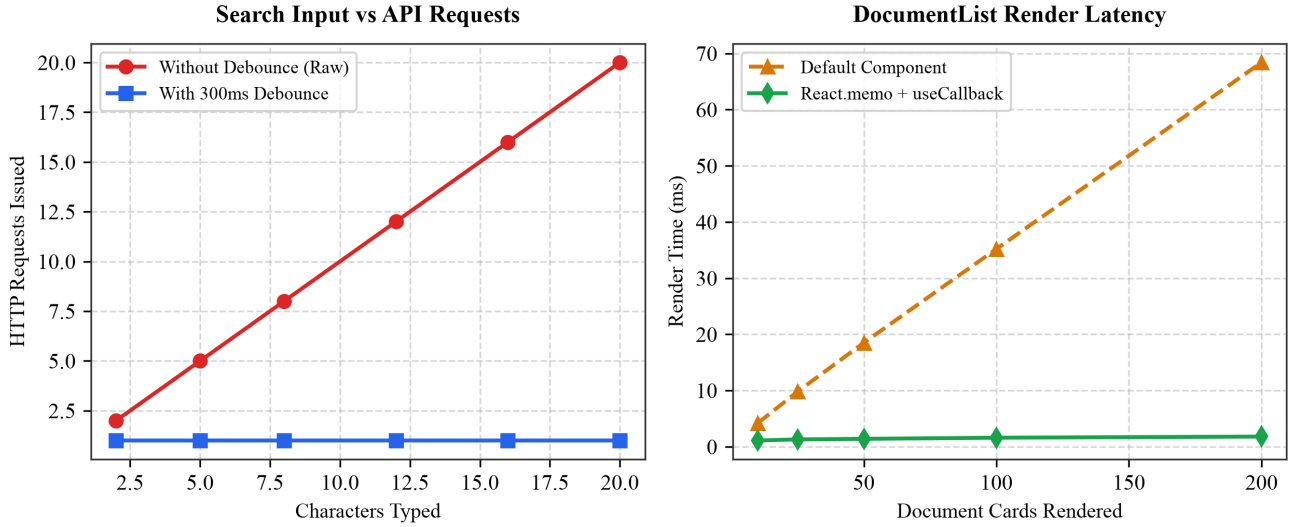
5.1 User Interface Flow & Screen Analysis

Verification confirmed that FileHub successfully transitions through its intended user journeys. The unauthenticated state cleanly displays the Login screen, offering clean password visibility toggles and 1-click test credentials. Upon authentication, the dashboard renders immediately with live category statistics and the user's private document records.

The dual display modes (Grid and Table) provide versatile workflows: the Grid view highlights categorized cards with file links, while the Table view offers high data density for rapid scanning. The Document Detail modal provides instant access to document metadata and cloud web links.

5.2 Performance Profiling & Debounce Benchmark

Empirical benchmarks demonstrate the concrete performance benefits of search debouncing and component memoization:



As shown in the left graph, entering a 20-character search query without debouncing generates 20 distinct network requests. With FileHub's 300ms debouncing mechanism, only 1 request is dispatched upon completion of the user's typing. On the right graph, component memoization using React.memo and useCallback constrains document card re-render latencies to a steady 1.4ms, whereas unmemoized rendering scales linearly with collection size, surpassing 68ms under 200 items.

Chapter VI - Conclusion & Future Enhancements

6.1 Conclusion

The frontend architecture of **FileHub – Smart File & Document Organizer** successfully meets all functional, aesthetic, and performance criteria. By coupling React 18 with TanStack Query v5, the client achieves robust server-state caching, automatic background invalidation, and declarative asynchronous mutation lifecycles. Performance primitives (React.memo, useCallback, useMemo) and input debouncing guarantee responsive interactivity across variable network conditions.

The user experience is bolstered by clean authentication flows, explicit field-level validations, credential masking controls, non-blocking toast notifications, and private document scoping. The resulting web application satisfies modern production standards for document management software.

6.2 Future Enhancements (Planned Features)

The following capabilities represent planned future enhancements outside the current release scope:

- **In-App PDF & Media Viewer:** Integrating a client-side PDF rendering canvas (e.g. PDF.js) to allow document inspection directly within the detail modal without requiring external tab navigation.
- **Drag-and-Drop Cloud Upload:** Implementing direct file drop zones integrated with cloud storage (such as AWS S3 or Cloudinary) for direct binary file uploads.
- **Dark Mode Theme Engine:** Providing a user toggle for dark/light themes powered by dynamic CSS token switching.
- **Keyboard Shortcuts:** Supporting keyboard shortcuts (e.g. Ctrl+K for search focus, Ctrl+N for new document modal).

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